Sahaj Bhadja

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Summary

Software engineer with 2+ years building automated test workflows, cloud services, and reliable distributed systems using Python, C++, PyTest, Mockito, Jenkins, AWS, Azure, Docker, and systematic defect isolation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Built fault-tolerant FastAPI workflows processing 5,000+ monthly jobs with explicit states, validation, reconciliation, failure isolation, and monitored releases, achieving 99%+ reliability across recurring production operations.
- Architected Azure SQL and Blob Storage services for 500+ users, analyzing operational metrics and tuning indexed access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed concurrent batch testing for 1,000+ uploaded files using multithreading, secure ingestion, AWS RDS, and fault-tolerant execution, isolating failures and improving throughput across production grading workflows.
- Implemented automated validation and deployment workflows for Flask services handling 10,000+ daily requests, reducing feedback turnaround by 60% while maintaining stable performance during peak test loads.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Automated nightly Jenkins pipelines for builds, VM installation, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, producing coverage summaries, CSV reports, and actionable failure logs.
- Troubleshot API and ETL defects using AWS Glue, S3, and SQL, improving response times by 30%, pipeline efficiency by 68%, and maintainability across distributed Agile engineering teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular MERN application with REST APIs, WebSockets, and eight services supporting 2M+ synchronized records while translating defined requirements into responsive customer and inventory workflows reliably.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed workflows, validation controls, and clear operational state tracking.
- Implemented webhook-based payment synchronization and root-cause troubleshooting, reducing manual handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

Skills

Testing & Quality: PyTest, Mockito, unit testing, integration testing, regression testing, test automation, test-case design, validation, defect tracking, root-cause analysis, failure isolation, quality assurance
Programming: Python, C++, C, Java, Bash, JavaScript, TypeScript, SQL, C#, PowerShell
Cloud & DevOps: Azure, AWS, Amazon S3, AWS Glue, AWS RDS, GCP, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes, Git, monitoring
Systems & Methods: Linux, Windows, macOS, REST APIs, FastAPI, Flask, microservices, distributed workflows, operational metrics, debugging, Agile/Scrum, code reviews, performance optimization, networking fundamentals

Projects

CPU-Scheduling Simulator Live | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% through deterministic simulation and systematic edge-case testing.

Linguistix | *Python, AWS ECS/Fargate, Docker, CodePipeline, CI/CD*
- Developed a containerized data-processing API on AWS with automated CodePipeline releases, executable validation checks, and Docker builds, achieving 87% accuracy and production-grade reliability across repeatable integration environments.

Blockchain Authentication Paper | *Security, IoT, Access Control*
- Evaluated secure IoT device authentication and role-based access across four permission tiers, designing decentralized verification flows that reduced repeated key-verification steps by 50% across modeled security scenarios.